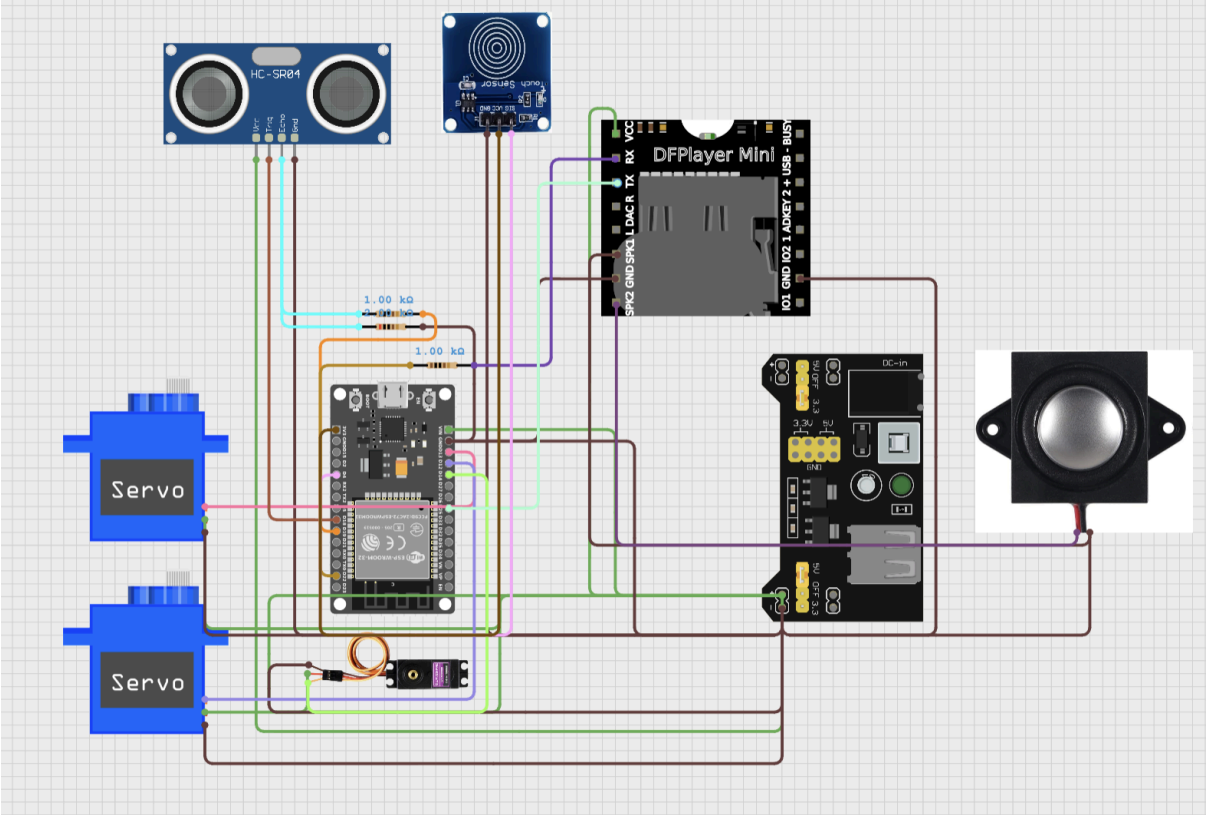
Electronics Documentation

1. Electronics Block Diagram



Power Flow

- External Power Supply (5V–6V)
  → Powers **MG996R servo (high current)**
  → Powers **micro servos**
  → Powers **DFPlayer Mini**
- ESP32 powered via:
  → USB / regulated 5V module
- **Common Ground**
  → All components share GND

System Logic Flow

Input → ESP32 → Decision → Action:

- Touch → flap wings + sound
- Motion → head movement + sound
- Idle → subtle movement + occasional sound

2. Components Table

Component	Quantity	Purpose	Justification
ESP32	*1*	*Main controller*	*Chosen for multiple GPIO pins, PWM support for servos, UART for DFPlayer, and better performance than Arduino for handling multiple sensors + behaviors simultaneously*
MG996R Servo Motor	*1*	*Wing flapping (ornithopter)*	*High torque servo required to drive mechanical wing system; smaller servos cannot handle load or resistance of flapping mechanism*
Micro Servo Motors	*2*	*Head movement (yaw + tilt)*	*Lightweight and precise, ideal for small angular movements needed for expressive head motion without adding excess weight*
HC-SR04 Ultrasonic Sensors	*2*	*Motion/distance detection*	*Allows directional sensing (left vs right), enabling the bird to orient toward the closest object, making behavior feel interactive*
TTP223 Touch Sensor	*1*	*User interaction trigger*	*Simple capacitive input that allows intuitive human interaction without mechanical switches, improving user experience*
DFPlayer Mini	*1*	*Audio playback*	*Standalone MP3 module with SD card support, reduces processing load on ESP32 and simplifies audio playback*
8 Ohm Speaker	*1*	*Sound output*	*Compatible with DFPlayer output, provides sufficient volume for small animatronic setup*
External Power Supply Module	*1*	*Power distribution*	*Necessary because servos (especially MG996R) draw high current that ESP32 cannot supply safely*